

Hospital Readmission Prediction Using Machine Learning

Submitted by

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Submission Date: 19 July 2026

Declaration

I hereby declare that this project titled “**Hospital Readmission Prediction Using Machine Learning**” is my original work carried out under the guidance of the academic and technical team of Denvey EduGrow. This project has not been submitted elsewhere for any academic degree or certification.

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Acknowledgement

I would like to express my sincere gratitude to **Denvey EduGrow** for providing the opportunity to work on this project. I am thankful to the mentors and instructors for their guidance, support, and encouragement throughout the project duration.

Abstract

Hospital readmissions within 30 days pose significant challenges to healthcare systems, leading to increased costs and reduced quality of care. This project presents a **machine learning-based hospital readmission prediction system** that analyzes patient demographic and clinical data to predict the likelihood of readmission. Feature engineering and data preprocessing techniques were applied before training a Random Forest classifier. The model was evaluated using standard classification metrics and demonstrated good predictive performance, highlighting the potential of machine learning in healthcare analytics.

1. Introduction

Hospital readmission is an important indicator of healthcare quality. Predicting patient readmission in advance allows healthcare providers to implement preventive strategies and improve patient care. With the growing availability of healthcare data, machine learning techniques can be effectively used to analyze patient records and support clinical decision-making.

2. Problem Statement

Manual analysis of patient medical records to predict readmission risk is inefficient and prone to error. There is a need for an automated system that can accurately predict hospital readmissions using historical patient data.

3. Objectives

- To analyze hospital patient data
- To predict 30-day readmission risk
- To preprocess and engineer meaningful features
- To evaluate the performance of the prediction model

4. Dataset Description

The dataset consists of hospital patient records containing demographic, medical, and hospital stay information. The dataset includes attributes such as age, gender, blood pressure, cholesterol, BMI, diabetes status, hypertension status, medication count, length of hospital stay, and discharge destination.

The target variable is **readmitted_30_days**, which indicates whether a patient was readmitted within 30 days.

5. Methodology

The methodology followed in this project includes:

1. **Data Collection:** Loading patient data from CSV files
2. **Data Preprocessing:** Handling categorical variables and feature encoding
3. **Feature Engineering:** Splitting blood pressure into systolic and diastolic values
4. **Model Training:** Training a Random Forest classifier
5. **Model Evaluation:** Evaluating performance using accuracy, precision, recall, and F1-score

6. Tools & Technologies Used

- Programming Language: Python
- Libraries: Pandas, Scikit-learn
- Development Environment: VS Code
- Version Control: GitHub

7. Machine Learning Model Used

- **Random Forest Classifier**

Random Forest was selected due to its ability to handle mixed data types and reduce overfitting through ensemble learning.

8. Performance Evaluation

The model performance was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score

The trained model achieved good predictive accuracy, indicating its effectiveness in identifying patients at risk of hospital readmission.

9. Results and Discussion

The results demonstrate that machine learning models can successfully predict hospital readmissions using patient data. The Random Forest classifier provided reliable predictions and handled feature interactions effectively. The model's performance can be further improved with larger datasets and advanced algorithms.

10. Conclusion

This project successfully demonstrates the application of machine learning techniques for hospital readmission prediction. The developed system can assist healthcare providers in identifying high-risk patients and improving overall healthcare outcomes.

11. Future Scope

- Incorporation of deep learning models
- Real-time prediction systems
- Integration with hospital information systems
- Deployment as a web-based healthcare application

12. References

1. Healthcare Analytics Datasets
2. Scikit-learn Documentation
3. Research papers on hospital readmission prediction